SRINIVAS UNIVERSITY
INSTITUTE OF ENGINEERING & TECHNOLOGY
MUKKA, MANGALURU-574146

IBM REPORT
ON

"FOOD SHARING APP"

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING

Submitted By,

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MUKKA, MANGALURU-574146

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Ibm report entitled ***"Food Sharing App"*** is a bonafide work carried out by ***Hotur Gargeya Datta, Guruprasad S Naik bearing the USN 01SU23CS077, 01SU23CS070*** in the partial fulfillment for the award of **Bachelor of Technology** in **Computer Science and Engineering** of the **Srinivas University-Institute of Engineering & Technology** during the year **2025-26.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The case study report has been approved as it satisfies the academic requirements in respect of Ibm report work prescribed for the said degree.

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SRINIVAS UNIVERSITY

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Department of Computer Science and Engineering

DECLARATION

We the students of Sixth semester, **B.Tech** in Computer Science and Engineering, Srinivas University, Mukka, hereby declare that the Project entitled **"*Food Sharing App*"** has been successfully completed by me in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology** in **Computer Science and Engineering of Srinivas University- Institute of Engineering & Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

-Hotur Gargeya Datta
-Guruprasad S Naik

Date:24-03-2026

Place: Mukka

ABSTRACT

Food wastage has become a critical global issue, contributing to environmental degradation and economic loss, while millions of people continue to suffer from hunger and malnutrition. The Food Sharing App is a web-based platform developed to bridge the gap between food donors and recipients by enabling the efficient redistribution of surplus food.

The application allows users such as households, restaurants, and event organizers to post details about excess food, including quantity, type, and location. Interested receivers, including NGOs and individuals, can browse available food and request it in real time. The system ensures quick communication, reduces food spoilage, and improves accessibility for needy people.

The application is developed using modern web technologies such as HTML, CSS, JavaScript for the frontend, and Node.js with MongoDB for backend and database management. The system is designed to be user-friendly, scalable, and efficient. By promoting food sharing and minimizing waste, this project contributes to social welfare and sustainable development.

The Food Sharing App is a full-stack web application designed to minimize food wastage and improve food distribution efficiency. The system provides a centralized platform where donors can upload surplus food information and receivers can access and claim available food resources.

The application follows a client-server architecture, where the frontend is developed using HTML, CSS, and JavaScript, and the backend is implemented using Node.js and Express.js. MongoDB is used as a NoSQL database to store user details, food listings, and transaction records. RESTful APIs are used for communication between client and server.

The system incorporates key functionalities such as user authentication, food posting, request handling, and database management. This project demonstrates the practical implementation of web technologies in solving real-world problems and promotes sustainable food management practices.

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-Hotur Gargeya Datta
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CHAPTER 1. INTRODUCTION

Food is one of the most essential resources for human survival, yet a significant amount of food is wasted every day across the world. Food wastage occurs at various levels such as households, restaurants, hotels, and large social events. At the same time, a large population continues to suffer from hunger, malnutrition, and lack of access to basic food resources. This imbalance highlights the urgent need for an efficient system to redistribute surplus food.

The Food Sharing App is designed as a digital solution to address this issue by connecting food donors with individuals or organizations in need. The application provides a platform where users can easily upload details about excess food, including type, quantity, location, and availability time. On the other hand, receivers such as NGOs or needy individuals can browse the available listings and request food accordingly.

With the advancement of web technologies, it has become easier to build platforms that enable real-time communication and coordination. This project leverages technologies such as HTML, CSS, and JavaScript for the frontend interface, along with Node.js and MongoDB for backend processing and data storage. The system ensures that the process of food donation and collection is simple, fast, and efficient.

The application also aims to promote social responsibility among users by encouraging them to contribute to reducing food waste. By providing an easy-to-use platform, the system motivates individuals and organizations to participate in food sharing activities. Additionally, it helps in reducing environmental impact, as food waste is a major contributor to pollution and greenhouse gas emissions.

Furthermore, the Food Sharing App supports community development by creating a network of donors and receivers. It ensures that surplus food is utilized effectively instead of being discarded. The system can be further enhanced with advanced features such as real-time notifications, location tracking, and integration with NGOs for better coordination.

In conclusion, this project is an initiative towards building a sustainable and socially responsible system that minimizes food wastage and helps in feeding the needy. It demonstrates how technology can be used effectively to solve real-world problems and improve the quality of life in society.

1.1. OBJECTIVES OF THE STUDY

- To reduce food wastage
- To provide food to needy people
- To create a user-friendly platform for food donation
- To connect donors and receivers efficiently
- To promote social responsibility

CHAPTER 2. PROBLEM STATEMENT

Food wastage is a major issue in modern society, occurring at various levels such as households, restaurants, hotels, supermarkets, and large social events. Every day, a significant amount of edible food is discarded due to over-preparation, poor planning, or lack of proper storage and distribution systems. At the same time, a large number of people suffer from hunger, malnutrition, and food insecurity. This imbalance highlights a critical gap between food availability and food accessibility.

One of the primary reasons for this issue is the absence of an efficient platform that connects food donors with those in need. Existing food donation practices are mostly manual, unorganized, and dependent on personal networks or non-governmental organizations. These methods are often time-consuming, lack real-time communication, and result in delays, which can lead to food spoilage and wastage.

Additionally, there is no centralized system to manage and track surplus food distribution. Donors often do not know where to donate, and receivers are unaware of available food resources nearby. This lack of coordination reduces the effectiveness of food donation efforts.

Another major concern is the environmental impact of food waste. Discarded food contributes to increased landfill waste and greenhouse gas emissions, which negatively affect the environment. Therefore, reducing food waste is not only a social necessity but also an environmental responsibility.

Furthermore, many existing solutions lack user-friendly interfaces and accessibility, limiting their adoption among common users. There is a need for a simple, efficient, and scalable system that allows users to easily share and access surplus food.

Hence, the problem addressed in this project is to design and develop a Food Sharing App that provides a digital platform for connecting food donors with receivers in real time. The system aims to reduce food wastage, improve food distribution efficiency, and contribute to social welfare by ensuring that surplus food reaches those who need it the most.

CHAPTER 3. SYSTEM ANALYSIS AND DESIGN

3.1 SYSTEM ANALYSIS

System analysis is the process of studying the existing system, identifying problems, and defining requirements for the proposed system.

3.1.1 Existing System

Currently, food donation is mostly handled manually through NGOs or personal contacts.

Limitations of Existing System

- Lack of proper communication between donors and receivers
- Time-consuming and inefficient process
- No real-time updates on food availability
- High chances of food spoilage
- Limited reach and accessibility
- No proper tracking or record maintenance

3.1.2 Proposed System

The proposed system is a web-based Food Sharing App that provides a digital platform for food donation and distribution.

Key Features

- Online food posting by donors
- Easy access for receivers to view available food
- Real-time updates and requests
- Secure login and user management
- Database for storing records

3.2 SYSTEM DESIGN

System design defines how the system will be implemented.

3.2.1 System Architecture

The application follows a three-tier architecture:

1. Presentation Layer (Frontend)
    - Built using HTML, CSS, JavaScript
    - Provides user interface
2. **Application Layer (Backend)**
    - Node.js with Express.js
    - Handles business logic and APIs
3. **Data Layer (Database)**
    - MongoDB
    - Stores user and food data

3.2.2 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

- User → Food Sharing System → Database
- System processes user requests and returns responses

Level 1 DFD

- User registration/login process
- Food posting and storage
- Food request and response handling

CHAPTER 4. METHODOLOGY

The development of the Food Sharing App follows a systematic and structured approach to ensure efficiency, reliability, and ease of use. The methodology adopted in this project is based on a step-by-step software development process.

4.1 Requirement Gathering

In this phase, the requirements of the system are identified and analyzed. Information is collected about:

- Types of users (donors, receivers, admin)
- Functional requirements (food posting, requesting, login system)
- Non-functional requirements (performance, usability, security)

This step ensures that the system meets user needs effectively.

4.2 System Design

The system design phase focuses on planning the structure and architecture of the application.

- Frontend Design: User interface layout using HTML, CSS, and JavaScript
- Backend Design: API structure using Node.js and Express.js
- Database Design: MongoDB collections for storing user and food data

Flowcharts, Data Flow Diagrams (DFD), and ER diagrams are created to represent system flow.

4.3 Development Approach

The project follows a modular development approach, where the system is divided into smaller modules:

- User Authentication Module
- Food Donation Module
- Food Request Module
- Admin Management Module

Each module is developed and tested separately before integration.

4.4 Frontend Development

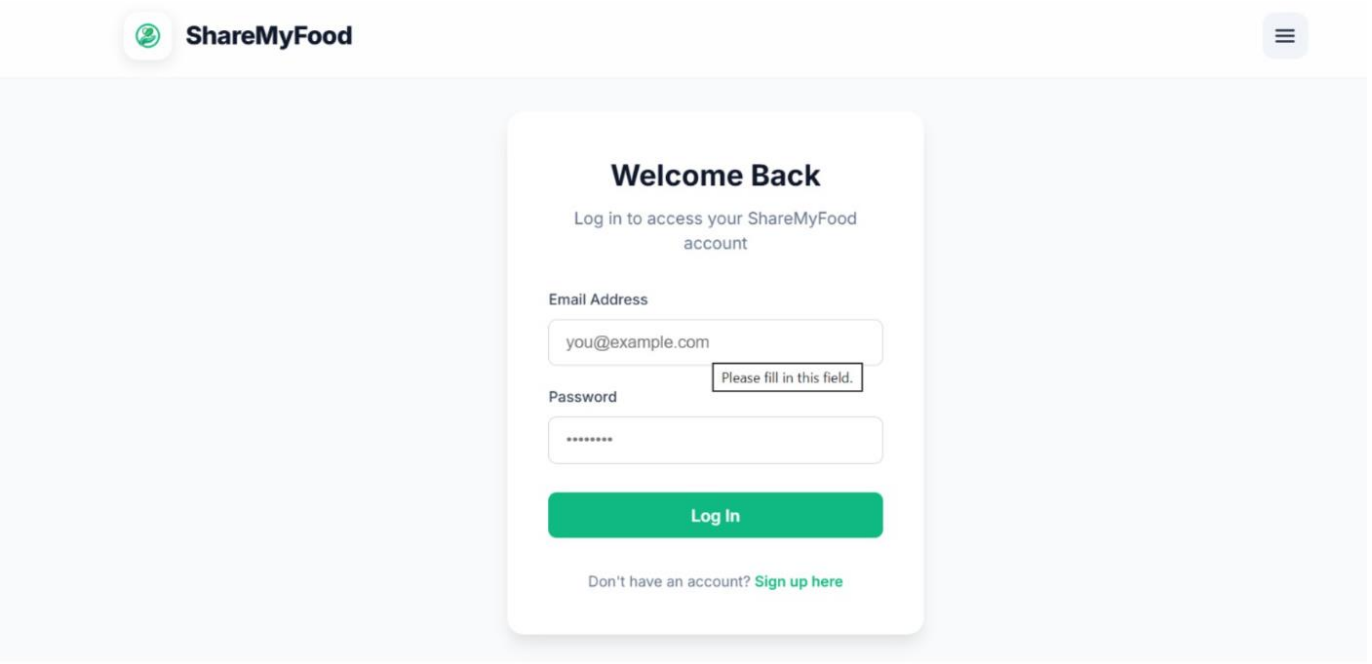
The frontend is developed using:

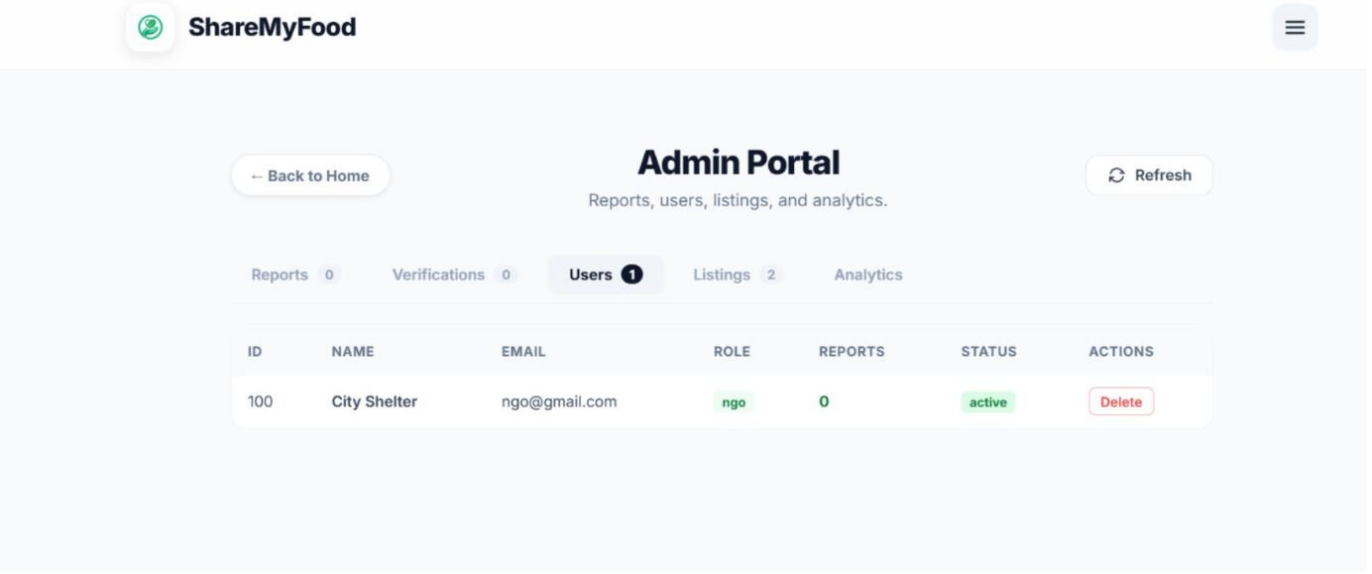
*HTML for structure
*CSS for styling
*JavaScript for interactivity

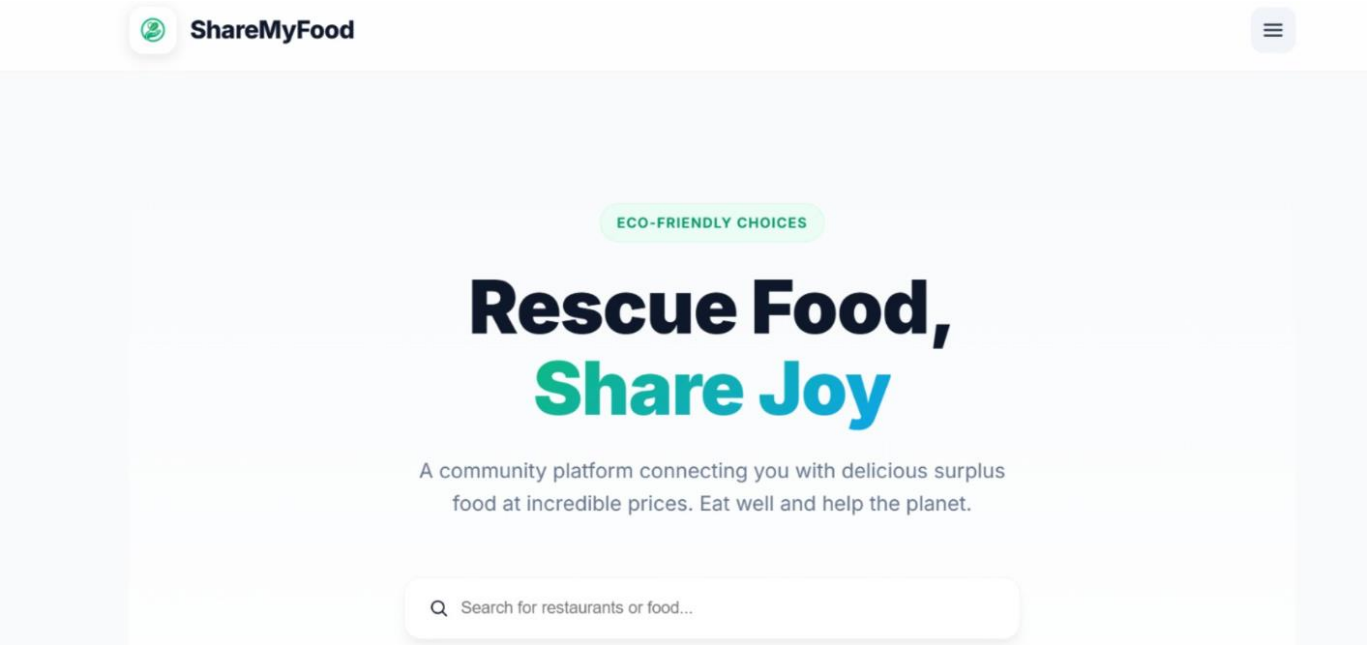
Features implemented:

- Registration and login forms
- Food listing interface
- Search and filter options

CHAPTER 5. O u t p u t






ShareMyFood
ECO-FRIENDLY CHOICES
Rescue Food,
Share Joy
A community platform connecting you with delicious surplus food at incredible prices. Eat well and help the planet.
Search for restaurants or food...

CHAPTER 6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The Food Sharing App is an effective solution designed to address the critical issues of food wastage and hunger in society. This project successfully demonstrates how technology can be used to bridge the gap between food donors and receivers by providing a simple and efficient digital platform.

The application enables users to easily donate surplus food and allows needy individuals or organizations to access it in real time. By integrating frontend technologies like HTML, CSS, and JavaScript with backend technologies such as Node.js and MongoDB, the system ensures smooth functionality, data management, and user interaction.

Through this project, the objectives of reducing food wastage, improving food distribution, and promoting social responsibility have been achieved. The system is user-friendly, scalable, and capable of handling multiple users efficiently. It also contributes to environmental sustainability by minimizing food disposal and reducing waste-related pollution.

In conclusion, the Food Sharing App is a practical and impactful solution that not only utilizes modern web technologies but also serves a meaningful social purpose. With proper implementation and wider adoption, it has the potential to make a significant difference in reducing hunger and food waste.

6.2 FUTURE ENHANCEMENTS

Although the current system is functional, several improvements can be made to enhance its performance and usability:

1. Mobile Application Development

- Develop Android/iOS app for better accessibility
- Improve user experience with mobile-friendly features

2. GPS and Location Tracking

- Real-time location tracking of donors and receivers
- Map-based food availability display

3. Notification System

- Push notifications for new food postings
- Alerts for request approvals and updates

4. AI-Based Recommendations

- Suggest nearby food based on user location
- Predict demand and optimize food distribution

5. NGO Integration

- Collaborate with NGOs for large-scale food distribution
- Automated coordination between donors and organizations

6. Rating and Feedback System

- Users can rate donors/receivers
- Improve trust and reliability

7. Food Quality and Safety Monitoring

- Add expiry alerts
- Guidelines for safe food handling

8. Multilingual Support

- Support multiple languages for wider reach

9. Payment/Donation Support

- Optional feature for monetary donations
- Support NGOs financially

10. Cloud Deployment

- Host on cloud platforms for scalability
- Improve performance and availability

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Research Papers / Articles

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3. FAO (Food and Agriculture Organization) Reports on Food Waste

Online Tutorials / Learning Platforms

1. freeCodeCamp – https://www.freecodecamp.org
2. GeeksforGeeks – https://www.geeksforgeeks.org
3. TutorialsPoint – https://www.tutorialspoint.com